time sticking together if there's something else that they can stick to first. So an impurity can set off a chain reaction where otherwise things might have continued on as they were. It turns out tiny black holes can be those nucleation sites for bubbles of true vacuum, but only if they're very small.

Fortunately for the universe, tiny black holes are not easy to make, given our current understanding of gravitational physics. Generally speaking, we only expect black holes to form at masses greater than that of the Sun, as the result of the collapse of massive stars at the end of their lives. Those black holes might grow to much larger masses by pulling in material or by merging with each other, but shrinking is another matter entirely. They can only lose mass via Hawking evaporation (see Chapter 4), and that takes ages. A black hole as massive as the Sun has an expected lifetime somewhere in the vicinity of 10^{64} years. At some point toward the end of that time, the black hole might get small enough to trigger vacuum decay, but we have quite a while before we really have to worry about that. It's also been hypothesized that in the early universe, tiny black holes might have formed due to the extreme densities of the Hot Big Bang, but so far we don't have any evidence of those. If they did form, though, and if little black holes really could destabilize the vacuum, we wouldn't be here. So if we take that into account, and believe in the possibility of vacuum decay, any theory that predicts tiny primordial black holes has to be wrong, because we exist.

Just for fun, a few of us have also been wondering whether there are ways to make those little black holes without them having been around since the very beginning of the universe. Making tiny black holes isn't a new idea. In addition to being awfully cute in a terrifying theoretical kind of way, these mini monsters could teach us about how gravity works, whether or not black holes really do that cool evaporation thing, and even whether or not there might be additional dimensions of space that we can't otherwise see.

For years, physicists have been combing the data from particle colliders, hoping to see some telltale sign that one of the collisions between protons managed to put so much energy into such a small space that it all immediately collapsed into a microscopic black hole. That black hole, if it appears, *should* be harmless, according to the traditional thinking, not taking the possibility of vacuum decay into account. Theory states that it should immediately evaporate via Hawking radiation, and that even if it